Case Study 01

Japan Rental Housing Project (2005)

Project Overview

Project Type
Rental Apartment

Location
Japan

Year
2005

Structure
Reinforced Concrete (RC)

Application
Adaptive Zero-Rebuild Asset System (AZRAS)

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1. Problem

Conventional apartment buildings are typically designed as fixed assets. As tenant needs change over time, major renovations often require demolition of interior walls, replacement of building services, and significant construction waste. These renovations increase lifecycle costs and shorten the effective economic life of the building.

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2. Conventional Design

In a conventional apartment building, structural components and interior systems are often integrated. Renovation therefore requires demolition work, increased labor, and interruption to occupancy.

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3. AZRAS Solution

The project adopted the AZRAS concept by separating the long-life structural core from replaceable interior systems. Exterior insulation and concrete thermal storage were incorporated to improve energy performance while allowing future interior modifications without rebuilding the structural frame.

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4. Results

Item	Result
Structural durability	Long-life RC structure
Interior adaptability	High
Future renovation	Simplified
Demolition waste	Reduced
Energy performance	Improved
Asset longevity	Enhanced

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5. Economic Perspective

Evaluation	Effect
Construction cost	Comparable to conventional methods*
Lifecycle cost	Reduced
Renovation cost	Reduced
Asset value	Better long-term retention

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6. Lessons Learned

This project demonstrated that separating structural durability from interior flexibility enables buildings to remain economically valuable for much longer than conventional construction. The experience later became one of the foundations of the AZRAS philosophy.

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7. Key Takeaways

✓ No structural demolition required
✓ Flexible interior renovation
✓ Long-life asset strategy
✓ Lower lifecycle cost
✓ Foundation of the AZRAS concept